



Homework assignment

- The purpose of this problem is to confirm the inapplicability of the square law formula for the 180 nm MOS devices in the ET4252 technology. Neglect finite output resistance in all calculations.
 - (a) Use an operating point simulation (.op) to measure the drain current of an NMOS transistor with $W/L = 5 \mu\text{m}/0.18 \mu\text{m}$ at $V_{DS} = 1.5\text{V}$ and $V_{GS} = 0.6\text{V}$ and 1.5V .
 - (b) Using the data from part (a), estimate the threshold voltage and $\mu_n C_{ox}(W/L)$ assuming the square law model. Note that the two unknowns, V_t and $\mu_n C_{ox}(W/L)$, can be calculated from the two simulated data points.
 - (c) Using the results from part (b) calculate the drain current and transconductance of an NMOS with an aspect ratio of $20\mu\text{m}/0.7\mu\text{m}$ at $V_{DS} = 1.5\text{V}$ and $V_{GS} = 0.75\text{V}$ using the square law formula. Next, use HSpice to find the actual current and transconductance and calculate the relative errors in percent. Note that percent errors are always calculated as $(\text{estimate} - \text{ideal value})/(\text{ideal value})$. What can you conclude about the validity of the square law model for this device?
 - (d) Extract the same values from part (c) using the lookup tables provided for the ET4252 technology. Show how they match the Hspice simulation through a screenshot of the MATLAB results.



Resources for Homework and Project

- All files needed are in the Tools folder on Brightspace
 - **et4252_hspice.sp**: technology HSPICE model
 - **180nch.mat** and **180pch.mat**: lookup tables extracted from the HSPICE model
 - **look_up.m** and **look_upVGS.m**: Matlab functions to query the lookup tables
 - **test_lookup.m**: Matlab example script to get familiar with the gm/ID methodology framework
 - **tools_demo.avi**: a simple demo video of the tools